

Lab Assignment 10

MTH 308: Numerical Analysis and Scientific Computing-I

Department of Mathematics and Statistics, IIT Kanpur

2026 Even Semester: April 24, 2026

In this lab, you will implement two numerical integration methods—Gauss–Legendre quadrature and Clenshaw–Curtis quadrature—and study their accuracy for different types of functions. Please write a single script file named `Lab10LastnameRollNumber.m` and publish it as a PDF. Then submit the script `Lab10LastnameRollNumber.m` along with the corresponding published file `Lab10LastnameRollNumber.pdf` (for example, my submitted files would be `Lab10Biswas260102.m` and `Lab10Biswas260102.pdf`) to HELLO IITK.

Consider the following integrals on the interval $[-1, 1]$:

(a) $I_1 = \int_{-1}^1 |x|^3 dx$

(b) $I_2 = \int_{-1}^1 e^{-x^2} dx$

(c) $I_3 = \int_{-1}^1 \frac{1}{1+x^2} dx$

(d) $I_4 = \int_{-1}^1 x^{10} dx$

1. Compute the exact values of the integrals (a)–(d). For $N = 1, 2, \dots, 50$, implement the Clenshaw–Curtis and Gauss–Legendre quadrature rules and compute numerical approximations of all four integrals. For each N , compute the absolute error

$$E_N = |I_{\text{numerical}} - I_{\text{exact}}|.$$

2. Plot the error versus N for both methods using a semilogarithmic scale.
3. Based on your results, comment on the convergence behavior of the two methods. In particular, discuss the effect of smoothness of the integrand and explain the behavior observed for $|x|^3$.
4. Using the function x^{10} , determine the degree of exactness of the quadrature rules.